Architecture Justification

Alumni Mentorship & Mock Interview Platform

Architecture Selection:
We chose Layered Architecture for the Alumni Mentorship & Mock Interview Platform.

Architectural Choice

The system follows a Layered Architecture consisting of a Presentation Layer, Business/Application Layer, and Data Layer. The Presentation Layer provides the interface for students and alumni mentors, the Business Layer handles mentor matching, availability, session booking, calendar integration, and post-interview scorecards, while the Data Layer manages persistent user, availability, booking, and scorecard information.

Reason 1 – Clear Separation of Responsibilities

The platform contains several distinct functions, such as mentor recommendation, availability management, session booking, calendar integration, and scorecard management. Layered Architecture separates these responsibilities into appropriate components, making the system easier to understand, maintain, test, and modify without unnecessarily affecting other parts of the application.

Reason 2 – Suitable for the Platform's Workflow

The architecture naturally supports the platform's workflow from student/mentor interaction → mentor matching and availability → session booking → calendar/meeting integration → post-interview scorecard. Business rules such as availability validation and prevention of double-booking can remain within the Business Layer, while persistent information is handled by the Data Layer. This provides a structured implementation of the requirements identified in Lab 1 and organized into Jira Epics in Lab 2.

Security Advantage

Layered Architecture helps enforce separation between the user interface, business logic, and stored data. Access to sensitive information such as user profiles, availability, bookings, and scorecards can be controlled through the Business Layer rather than allowing the presentation layer to directly access the database. This supports the platform's privacy requirements and helps reduce unauthorized direct data access.

Performance Benefit

The separation of frequently used application operations from data storage allows the system to optimize each layer independently. For example, mentor recommendations and availability checks can be handled in the Business Layer while the Data Layer efficiently retrieves only the required profile, availability, and booking information. This can reduce unnecessary processing and improve response time for common operations such as viewing mentor availability and booking a session.